Stochastic Processes 2021/22 Semester 1

Tutorial 9 Extension Questions

Please note: These questions are an extension to Tutorial 9 for individual training only and are not supposed to be discussed during tutorials or office hours. Answers are available for all questions, but worked solutions are only available for some hard questions.

1. Redo questions (a) to (c) in Tutorial 9 Question 2, with the new transition rate matrix

$$Q = \begin{pmatrix} -4 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -3 & 0 & 1 & 1 \\ 0 & 2 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 3 \\ 0 & 0 & 0 & 0 & 2 & -2 \end{pmatrix}$$

2. Consider a Galton-Watson branching process with offspring distribution:

$$p_1 = p_2 = 1/2, \quad p_k = 0 \text{ for all other } k.$$

   (a) Find the expected time to reach 0 if the initial state is $X_0 = 1$.

   (b) Find the expected time to reach 2 if the initial state is $X_0 = 1$

3. Consider a Galton-Watson branching process with offspring distribution

$$p_1 = 2/5, p_2 = 3/5, \quad p_k = 0 \text{ for all other } k.$$

   Find the expected time
   (a) to hit the set $\{2, 3, 4, \ldots\}$ if $X_0 = 1$;
   (b) to hit the set $\{3, 4, 5, \ldots\}$ if $X_0 = 2$;
   (c) to hit the set $\{4, 5, 6, \ldots\}$ if $X_0 = 2$.